

Methods

A method contains a set of instructions that perform well-defined task and **only runs** when it is called. For example, calculating the area of a circle, displaying a particular message on the screen, etc.

When programmers create methods of their own choice to perform some task. This type of methods called **user-defined**. Another type is **built-in** methods that are readily available for use such as `System.out.println()`, `sqrt()`, `compareTo()` and `nextInt()`.

The benefits of using methods are:

1. Construct the program from simple, small pieces or components.
2. Avoid repeating code.
3. Software Reuse.

To declare a method in Java:

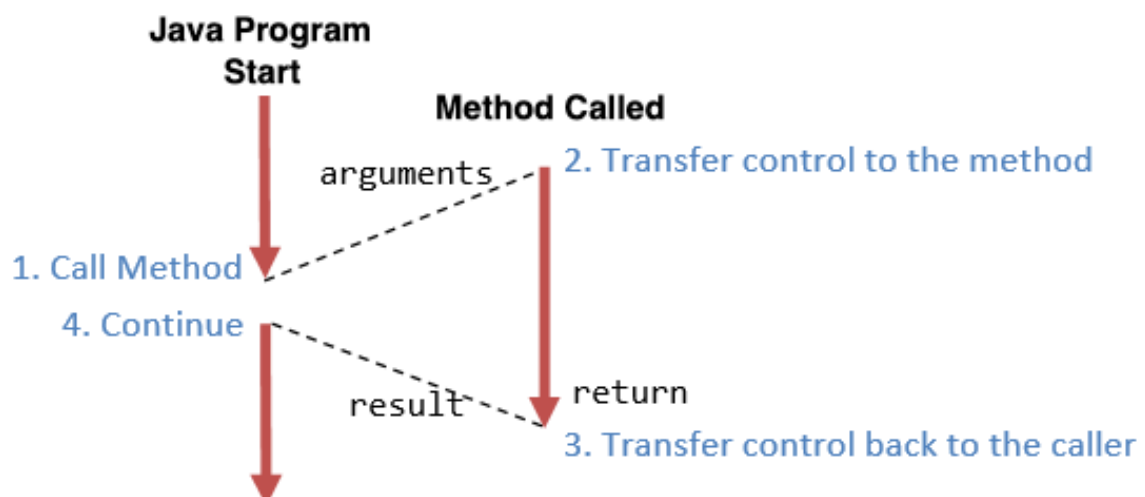
A Java method must always be a part of a class in Java.

Syntax:

Attributes Return_type Method_name (Parameters)

```
{  
    Statement1;  
    Statement2;  
    .  
    .  
    .  
}
```

Body of the method



Adding a method:

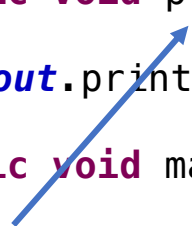
```
public static void NAME()  
{  
    Statement(s);  
}
```

To call a method inside a Java program:

NAME();

Example1:

```
public class MethodExample1 {  
    public static void printmethod()  
    {  
        System.out.println("Welcome to Java Programming!");  
    }  
    public static void main(String arg[])  
    {  
        printmethod();  
    }  
}
```

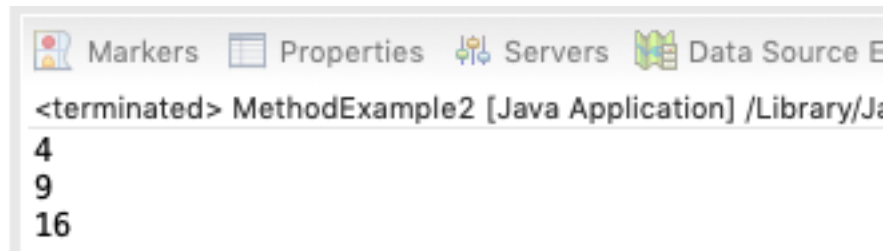


```
public class MethodExample1 {  
    public static void printmethod()  
    {  
        System.out.println("Welcome to Java Programming!");  
    }  
    public static void main(String arg[])  
    {  
        printmethod();  
        printmethod();  
        printmethod();  
    }  
}
```



Example2:

```
public class MethodExample2 {  
    public static void main(String[] arguments) {  
        int value = 2;  
        printSquare(value);  
        printSquare(3);  
        printSquare(value*2);  
    }  
  
    public static void printSquare(int x) {  
        System.out.println(x*x);  
    }  
}
```



```
public class MethodExample2 {  
    public static void main(String[] arguments) {  
        printSquare("hello");  
        printSquare(3.5);  
    }  
  
    public static void printSquare(int x) {  
        System.out.println(x*x);  
    }  
}
```

What is wrong here?

Homework:

- Q1>** Write a Java program has a method that takes a string as input and print it out.
- Q2>** Write a Java program has a method that takes a double number as input and check whether it positive or negative.